

be done for all run-lengths r_t at each time step t for the recursive relationship powering the algorithm, this represents a substantive computational overhead.

Taken together, these issues often render posteriors based on the β -divergence computationally infeasible; especially in high-dimensional settings. In principle, one could replace the β -divergence with various robust alternatives whose numerical issues are less substantive and whose hyperparameters are easier to tune—such as α -divergences (Hooker & Vidyashankar, 2014), γ -divergences (Knoblauch, 2019), or maximum mean discrepancies (Chérif-Abdellatif & Alquier, 2020). Unfortunately, none of these alternatives alleviate the problem of computationally expensive variational approximations. This is an issue, since ultimately, it is the conjugate forms that can be updated in terms of sufficient statistics that make BOCD computationally attractive.

In the face of this, it may be tempting to postulate an inherent trade-off between robustness and computational tractability for generalised Bayes. But this is not so; recently, it was shown that robust posteriors based on kernel Stein discrepancies have a conjugacy property (Proposition 2 of Matsubara et al., 2022b). These generalised posteriors however are not suitable for BOCD: Updating them from $t - 1$ to t observations takes $\mathcal{O}(t)$ operations—as opposed to the $\mathcal{O}(1)$ operations required for standard Bayesian posteriors. Such updates would lead to an algorithm whose computational demands per iteration increase linearly the longer it is run, leading to an ‘online’ algorithm in name only. This is why the current paper proposes a new class of generalised posteriors based on diffusion score matching (Barp et al., 2019): we prove that they are robust, and lead to conjugacy, with closed forms updates that take $\mathcal{O}(1)$ operations.

3. Methodology

We present the methodological innovations of the current paper in three steps: After an exposition of diffusion score matching, we first explain how the resulting generalised Bayesian posterior yields closed form updates. In a second step, we provide formal robustness guarantees for these posteriors. In the last step, we show how to integrate them into the BOCD framework, yielding $\mathcal{D}_m$ -BOCD; and how to choose its hyperparameters.

3.1. Diffusion Score Matching Bayes

Notation. We write the divergence operator on a vector field f as $\nabla \cdot f$. This condenses the formulae derived in this paper, but we provide all uncondensed versions in Appendix A. The d -dimensional vector (and $d \times p$ sized matrix) of partial derivatives for $f : \mathcal{X} \rightarrow \mathbb{R}$ (and $g : \mathcal{X} \rightarrow \mathbb{R}^p$) evaluated at $x \in \mathcal{X}$ is written as $\nabla f(x)$ (and $\nabla g(x)$).

Score Matching. Score matching is a discrepancy-based method for estimating parameters first proposed by Hyvärinen (2006). The key idea is to approximately minimise the Fisher divergence between the statistical model $\{p_\theta : \theta \in \Theta\}$ and the data-generating process p_0 . This method takes its name from the fact that for a density p on $\mathcal{X}$ and $s_p(x) = \nabla \log p(x)$ —the so-called *score function* of the density p —the Fisher divergence is

$$\mathcal{D}_{I_d}(p_0 || p_\theta) = \mathbb{E}_{X \sim p_0} [\|s_{p_\theta}(X) - s_{p_0}(X)\|_2^2].$$

This divergence is therefore minimised by matching the scores of the model to that of the data-generating process p_0 . This objective is convenient for two main reasons: Firstly, for the density $p = \tilde{p} \frac{1}{Z}$ with normaliser $Z > 0$, $s_p = s_{\tilde{p}}$, so that the objective is attractive when working with likelihoods whose normaliser Z is unknown. Secondly, the objective can be rewritten so that the scores of p_0 do not have to be estimated to compute it.

Score matching has been used widely, including for data on manifolds or other complex domains (Mardia et al., 2016; Liu et al., 2022; Scealy & Wood, 2022), energy-based models (Vincent, 2011), anomaly detection (Zhai et al., 2016), nonparametric density estimation (Sriperumbudur et al., 2017), score-based generative modelling (Song & Ermon, 2019), and even for Bayesian model selection (Dawid & Musio, 2015; Shao et al., 2019; Jewson & Rossell, 2022) or as a scoring rule (Parry et al., 2012). In recent work, Wu et al. (2023) used score matching for change point detection. This work differs from ours in three major ways: they consider a frequentist setting based on the CUSUM statistic, they only consider standard score matching, and they are not concerned with robustness. Building on these successes, various generalised forms of score matching have been proposed over the years to address some of its shortcomings (e.g. Lyu, 2009; Xu et al., 2022; Yu et al., 2022; Matsubara et al., 2022a).

Diffusion Score Matching. The particular generalisation we consider hereafter is *diffusion score matching*, which was introduced in Barp et al. (2019) and amounts to a weighted version of the Fisher divergence given as

$$\mathcal{D}_m(p_0 || p_\theta) = \mathbb{E}_{X \sim p_0} [\|m^\top(X)(s_{p_\theta}(X) - s_{p_0}(X))\|_2^2],$$

for a pointwise invertible matrix-valued function $m : \mathcal{X} \rightarrow \mathbb{R}^{d \times d}$. The function m is also known as diffusion matrix due to the construction of this distance as a Stein discrepancy with a pre-conditioned diffusion Stein operator; see Anastasiou et al. (2023) for full details.

Like $\mathcal{D}_{I_d}$, $\mathcal{D}_m$ is a statistical divergence between densities p_0 and p_θ on $\mathcal{X} = \mathbb{R}^d$ whenever $\int_{\mathcal{X}} |s_{p_\theta}(x) - s_{p_0}(x)|^2 p_0(x) dx < \infty$. Under appropriate smoothness and boundary conditions, this can be extended to the case where $\mathcal{X}$ is a connected subset of $\mathbb{R}^d$ (Liu et al., 2022;

Zhang et al., 2022). More generally, $\mathcal{D}_m$ recovers $\mathcal{D}_{I_d}$ for $m(x) = I_d$ (the d -dimensional identity matrix), the estimator in Hyvärinen (2007) for $m(x) = x$, and the generalised h-score matching method for $m(x) = \text{diag}(h^{1/2}(x))$, where h is defined in Yu et al. (2018; 2019). The function m can be thought of as up-weighting areas of $\mathcal{X}$ on which matching the scores of the model to that of the data-generating process is most important. For the purposes of the current paper, we will choose this weight to ensure that the constructed generalised posteriors are provably robust (see Section 3.3 for details).

Estimating $\mathcal{D}_m$ directly is challenging, as it would require estimating the unknown score s_{p_0} . Fortunately, under the aforementioned smoothness and boundary conditions (Appendix B.5) (Liu et al., 2022), we can expand the above equation and use integration by parts. Then, up to a constant that does not depend on θ , we can rewrite $\mathcal{D}_m(p_0 \| p_\theta)$ as

$$\mathbb{E}_{X \sim p_0} [\|(m^\top s_{p_\theta})(X)\|_2^2 + (2\nabla \cdot (mm^\top s_{p_\theta}))(X)]. \quad (3)$$

Crucially, the quantity above no longer features s_{p_0} , and only depends on p_0 through an expectation. This leads to a natural estimator which for $x_{1:T}$ is given by

$$\begin{aligned} \hat{\mathcal{D}}_m(\theta) &= \frac{1}{T} \sum_{t=1}^T d_m(\theta, x_t), \quad \text{where} \\ d_m(\theta, x_t) &= \|(m^\top s_{p_\theta})(x_t)\|_2^2 + (2\nabla \cdot (mm^\top s_{p_\theta}))(x_t). \end{aligned}$$

Diffusion Score Matching Bayes. Based on the estimator $\hat{\mathcal{D}}_m$ for the part of $\mathcal{D}_m$ that depends on θ , we can construct

$$\pi_\omega^{\mathcal{D}_m}(\theta | x_{1:T}) \propto \pi(\theta) \exp(-\omega T \hat{\mathcal{D}}_m(\theta)). \quad (4)$$

Using score matching for a generalised Bayes posterior was first discussed in passing in Section 4.2 of Giummolè et al. (2019), though the context is about reference priors for objective Bayesian inference, and the method is only briefly mentioned. This previous work also does not robustify the resulting posterior through the introduction of a weighting matrix m , or derive its conjugate posteriors.

3.2. Conjugacy for Exponential Family Models

The conjugacy of posteriors of the form (4) make them more attractive than potential alternatives. For exponential family likelihoods, these posteriors depend on two parameters available in closed form. The exponential family is given the collection of models with a probability density function

$$p_\theta(x) = \exp(\eta(\theta)^\top r(x) - a(\theta) + b(x)), \quad (5)$$

where $\eta : \Theta \rightarrow \mathbb{R}^p$, $r : \mathcal{X} \rightarrow \mathbb{R}^p$, $a : \Theta \rightarrow \mathbb{R}$, and $b : \mathcal{X} \rightarrow \mathbb{R}$. When $\eta(\theta) = \theta$, we say that the exponential family model is in natural form, and one can reparametrise a model to natural form by reparameterising with the map η^{-1} . Exponential family class of distributions includes the Gaussian, exponential, Gamma, and Beta distributions.

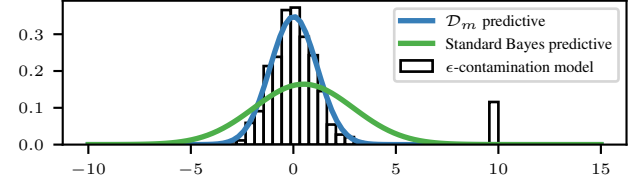


Figure 2. Impact of misspecification in posteriors. The robust $\mathcal{D}_m$ -posterior and non-robust standard Bayes posterior predictive when the data are incorrectly modelled as Gaussian, but follow an ε -contamination model $\mathbb{P} = 0.95\mathcal{N}(0, 1) + 0.05\delta_{10}$.

Proposition 3.1. If p_θ is given by (5), then

$$\pi_\omega^{\mathcal{D}_m}(\theta | x_{1:T}) \propto \pi(\theta) \exp(-\omega T [\eta(\theta)^\top \Lambda_T \eta(\theta) + \eta(\theta)^\top \nu_T]),$$

for $\Lambda_T = \frac{1}{T} \sum_{t=1}^T \Lambda(x_t)$, $\nu_T = \frac{2}{T} \sum_{t=1}^T \nu(x_t)$, and

$$\begin{aligned} \Lambda(x) &= (\nabla r^\top m m^\top \nabla r)(x), \\ \nu(x) &= (\nabla r^\top m m^\top \nabla b + \nabla \cdot (m m^\top \nabla r))(x). \end{aligned}$$

Taking $\eta(\theta) = \theta$ and choosing a squared exponential prior $\pi(\theta) \propto \exp(-\frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu))$, also makes $\pi_\omega^{\mathcal{D}_m}(\theta | x_{1:T})$ a (truncated) normal of the form

$$\pi_\omega^{\mathcal{D}_m}(\theta | x_{1:T}) \propto \exp(-\frac{1}{2}(\theta - \mu_T)^\top \Sigma_T^{-1}(\theta - \mu_T)),$$

for $\Sigma_T^{-1} = \Sigma^{-1} + 2\omega T \Lambda_T$ and $\mu_T = \Sigma_T (\Sigma^{-1} \mu - \omega T \nu_T)$.

The proof is in Appendix B.1. The natural exponential family allows us to recover a form of Gaussian conjugacy, since the diffusion score matching squared becomes a quadratic form in this case. This renders DSM-Bayes scalable; as we will elaborate upon in Section 3.4, Σ_T^{-1} and μ_T can be updated with a new observation in $\mathcal{O}(p^2 + d^2)$ operations.

3.3. Global Bias-Robustness

Building a BOCD algorithm based on $\pi_\omega^{\mathcal{D}_m}(\theta | x_{1:T})$ is attractive not only computationally, but also due to its robustness. We prove this robustness formally by using the classical framework of ε -contamination models (see, e.g. Huber, 1981). Given a distribution $\mathbb{P}$, we consider its ε -contaminated counterpart $\mathbb{P}_{\varepsilon, y} = (1 - \varepsilon)\mathbb{P} + \varepsilon\delta_y$, where δ_y is the dirac-measure at some $y \in \mathcal{X}$, and $\varepsilon \in [0, 1]$. The classical perspective on robustness proceeds by defining a point estimator $E : \mathcal{P}(\mathcal{X}) \rightarrow \Theta$ that maps from $\mathcal{P}(\mathcal{X})$, the space of distributions on $\mathcal{X}$, to Θ . One then investigates its robustness via $\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \|E(\mathbb{P}) - E(\mathbb{P}_{\varepsilon, y})\|_2$, which under mild conditions is equivalent to the derivative $\frac{\partial}{\partial \varepsilon} \|E(\mathbb{P}_{\varepsilon, y})\|_2 \big|_{\varepsilon=0}$. This limit is the so-called *influence function*. It quantifies the impact of an infinitesimal contamination at y on the estimator, and is a classical tool to measure outlier robustness.

The Bayesian case is slightly more complicated and depicted in Figure 2: we are not concerned by estimators on Θ , but on $\mathcal{P}(\Theta)$. The estimates under study are thus infinite-dimensional objects that vary over Θ . To get a

handle on this, we first define an influence function *pointwise* for each $\theta \in \Theta$. To this end, note that $\widehat{\mathcal{D}}_m(\theta) = \mathbb{E}_{X \sim \mathbb{P}_T}[d_m(\theta, X)]$. We can now define the density-valued estimator $\pi_{\omega}^{\mathcal{D}_m}(\theta|\mathbb{P}) \propto \pi(\theta) \exp\{-\omega T \mathbb{E}_{X \sim \mathbb{P}}[d_m(\theta, X)]\}$, noting $\pi_{\omega}^{\mathcal{D}_m}(\theta|\mathbb{P}_T) = \pi_{\omega}^{\mathcal{D}_m}(\theta|x_{1:T})$ for $\mathbb{P}_T = \frac{1}{T} \sum_{t=1}^T \delta_{x_t}$. Its pointwise posterior influence function (PIF) is

$$\text{PIF}(y, \theta, \mathbb{P}) = \frac{d}{d\varepsilon} \pi_{\omega}^{\mathcal{D}_m}(\theta|\mathbb{P}_{\varepsilon, y}) \Big|_{\varepsilon=0}.$$

Since this is a definition of sensitivity that is local to both θ and y , making a global statement for all of $\pi_{\omega}^{\mathcal{D}_m}(\theta|x_{1:T})$ requires that we aggregate a notion of sensitivity over both arguments. The easiest way to do this is to investigate $\sup_{\theta \in \Theta, y \in \mathcal{X}} \text{PIF}(y, \theta, \mathbb{P}_T)$. If this double supremum is bounded, we call a posterior *globally bias-robust*, which means that the impact of contamination on the posterior density is uniformly bounded—both over the parameter space, and the location of said contamination in the data space. This way of studying the robustness of generalised posteriors was pioneered in Ghosh & Basu (2016), and extended by Matsubara et al. (2022b). We build on these advances, and provide a simple condition on m for global bias-robustness of $\pi_{\omega}^{\mathcal{D}_m}(\theta|x_{1:T})$ in some exponential family models.

Proposition 3.2. *If p_{θ} is as in (5) so that $\eta(\theta) = \theta$ and $\nabla b = 0$, and if the prior is a squared exponential as in Proposition 3.1, then $\pi_{\omega}^{\mathcal{D}_m}(\theta|x_{1:T})$ is globally bias-robust if $m : \mathcal{X} \rightarrow \mathbb{R}^{d \times d}$ is chosen so that $\theta^* \neq 0_p$ and*

$$m_{ij}(x) = \begin{cases} \frac{1}{\sqrt{1 + (\nabla r(x)\theta^*)_i^2}} & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

While m could in principle depend on θ , this would break the conjugacy presented in Proposition 3.1. The above choice of m does *not* depend on θ , and therefore maintains the computational advantages of $\pi_{\omega}^{\mathcal{D}_m}(\theta|x_{1:T})$. The result’s conditions are also mild: we can always ensure that $\eta(\theta) = \theta$ by re-parameterising. Similarly, most distributions of interest satisfy $\nabla b = 0$. Examples include Gaussians, exponentials, (inverse) Gamma, and Beta distributions. Note also that m is only applicable to models with support $\mathcal{X} = \mathbb{R}^d$, as the expansion in (3) is otherwise not valid without additional boundary conditions. However, we prove that the proposed weight matrix m also leads to a well-defined discrepancy measure for various distributions defined on subsets of $\mathcal{X}$, including the Gamma and the exponential distribution (see Appendix B.5).

3.4. $\mathcal{D}_m$ -BOCD

Using our robust posterior within BOCD is straightforward, as its only appearance is in the posterior predictive via

$$p(x_t|x_{t-1}^{(r)}) = \int_{\Theta} p_{\theta}(x_t) \pi_{\omega}^{\mathcal{D}_m}(\theta|x_{t-1}^{(r)}) d\theta.$$

If p_{θ} is a natural exponential family with a squared exponential prior, then $\pi_{\omega}^{\mathcal{D}_m}(\theta|x_{t-1}^{(r)})$ is a normal distribution parameterised by inverse covariance matrix $\Sigma_{t-1,r}^{-1}$ and mean $\mu_{t-1,r}$ by virtue of 3.1. This makes the predictive easy to compute—either in closed form or by sampling from $\pi_{\omega}^{\mathcal{D}_m}$ —which is a significant advantage over the β -BOCD framework. For the latter, the posterior will generally be intractable so that the algorithm relies on variational approximations. Importantly, there is no way to both efficiently and exactly update variational approximations based on $x_{1:t}$ once observation x_{t+1} arrives: one either uses cheap updates that lead to subpar variational approximations of the posterior, or one re-computes the approximation from scratch at the expense of a substantive computational overhead.

In contrast, our approach allows for a cheap and exact update: if we store $\Sigma_{t-1,r}^{-1}$ and $\mu_{t-1,r}$, we can perform the update $\pi_{\omega}^{\mathcal{D}_m}(\theta|x_{t-1}^{(r)}) \mapsto \pi_{\omega}^{\mathcal{D}_m}(\theta|x_t^{(r+1)})$ that adds x_t into the parameter posterior of the segment $x_{t-1}^{(r)}$ via

$$\begin{aligned} \Sigma_{t,r+1}^{-1} &= \Sigma_{t-1,r}^{-1} + 2\omega\Lambda(x_t), \\ \mu_{t,r+1} &= \Sigma_{t,r+1} (\Sigma_{t-1,r}^{-1} \mu_{t-1,r} - 2\omega\nu(x_t)). \end{aligned}$$

If we have access to the un-inverted matrix $\Sigma_{t,r+1}$, all of these operations are basic matrix and vector additions or multiplications that take $\mathcal{O}(p^2 + d^2)$ operations to execute. While naively computing $\Sigma_{t,r+1}$ from $\Sigma_{t-1,r}^{-1}$ would take $\mathcal{O}(p^3)$ operations, we can apply the Sherman-Morrison formula to the update of $\Sigma_{t,r+1}^{-1}$ to reduce this to $\mathcal{O}(p^2)$, maintaining the overall complexity of $\mathcal{O}(p^2 + d^2)$. This is also the complexity of standard BOCD with the Gaussian likelihood and conjugate prior (Adams & MacKay, 2007). In CP methods for high-frequency data, both the number of parameters p and the data dimension d are typically small, so that an update of $\mathcal{O}(p^2 + d^2)$ is attractive.

Run-length pruning. A naive implementation of $\mathcal{D}_m$ -BOCD would keep a posterior over all possible run-lengths $r_t = \{0, 1, \dots, t-1\}$, but this would lead to an algorithm with overall complexity $\mathcal{O}(\sum_{t=1}^T t(d^2 + p^2)) = \mathcal{O}(T^2(d^2 + p^2))$ for a time series of length T . To prevent this, authors have proposed to ‘prune’ the run-length posterior to a constant length (Adams & MacKay, 2007; Fearnhead & Liu, 2007). Here, we follow the most popular strategy (e.g. Adams & MacKay, 2007; Saatçi et al., 2010; Knoblauch & Damoulas, 2018) by keeping only the k most probable run-lengths. For all experiments, we take $k = 50$.

Choice of m Throughout, we choose m as per Proposition 3.2, as it ensures robustness—even for certain distributions with boundaries (see Appendix B.5). Regarding θ^* , we found that $\mathcal{D}_m$ -BOCD was not very sensitive to this choice; likely because tuning ω offsets any sensitivity to it. In all experiments, we thus picked θ^* as the maximum likelihood estimate computed on the full data set. We note that one